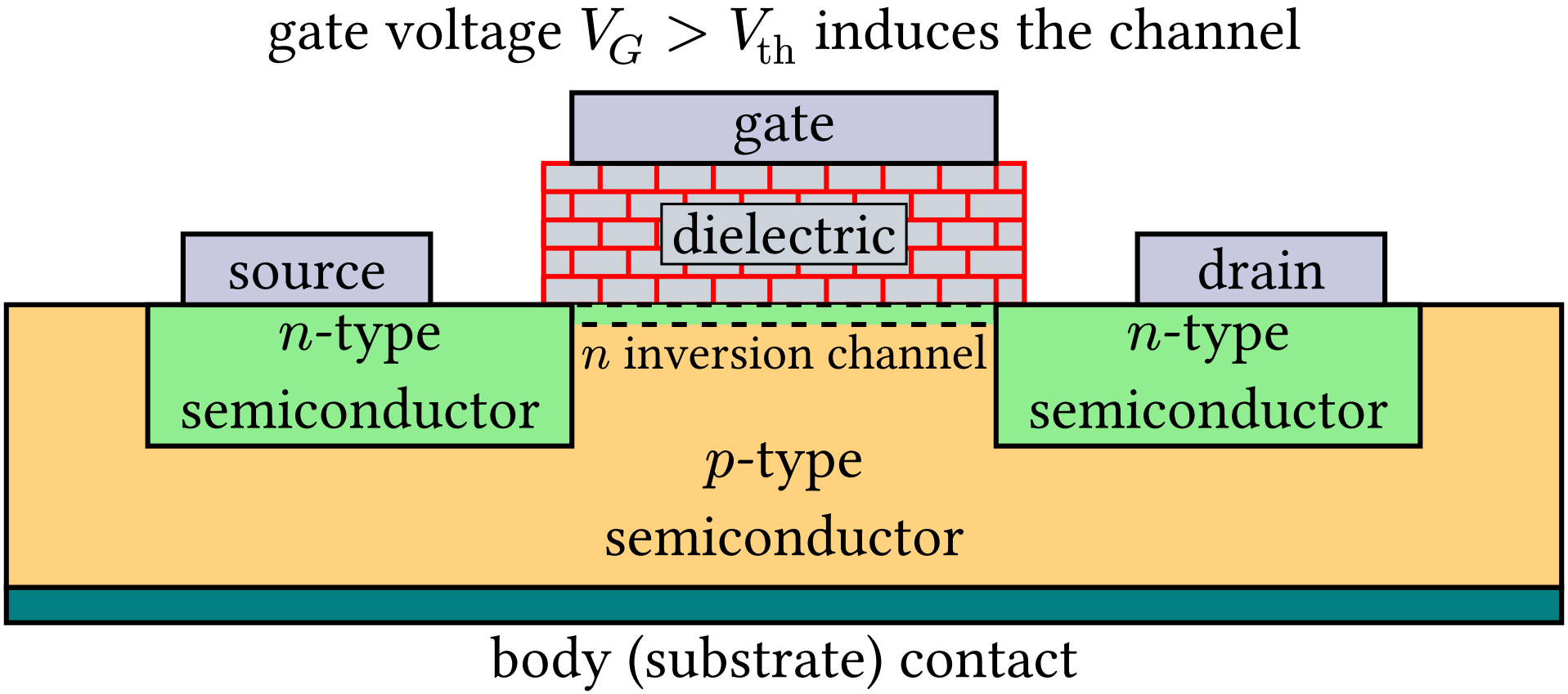
gate voltage $V_G > V_{\text{th}}$ induces the channel
gate
dielectric
source
drain
n-type
semiconductor
n inversion channel
n-type
semiconductor
p-type
semiconductor
body (substrate) contact